

SUPERSTORE RETAIL ANALYTICS

Project Documentation

Total Sales \$2.33M 5,111 orders	Total Profit \$292K Net positive	Profit Margin 12.6% Furniture drags this	Avg Order \$464 All segments
---	---	---	---

Prepared by: Gauransh Singh (Garv)

BCA Final Year | AI/ML Specialization | Galgotias University

Tools: MySQL 8.0 | Microsoft Excel | Power BI

Portfolio: gauransh-singh.github.io | LinkedIn: linkedin.com/in/gauransh-singh-211586294

1. Project Overview

This project performs a full-cycle retail analytics study on the Sample Superstore dataset, a widely used benchmark dataset in business intelligence. The goal is to identify where the business is generating value and where it is losing money, using SQL for structured analysis and Excel for interactive dashboard visualization.

1.1 Objectives

- Clean and validate raw transactional retail data
- Perform exploratory and advanced SQL analysis across 8 dimensions
- Identify loss-making products, sub-categories, and discount patterns
- Build an interactive Excel dashboard with KPIs, charts, and slicers
- Derive actionable business recommendations from data findings

1.2 Business Question

| *"Where is the Superstore bleeding money, and what is causing it?"*

Every analysis phase feeds this central question. Time analysis shows when, regional analysis shows where, RFM shows who, discount banding shows why, and sub-category margin shows what.

2. Dataset

2.1 Source

- Name: Sample Superstore Dataset
- Format: CSV (samplesuperstore.csv)
- Rows: 10,194 records (10,192 after duplicate removal)
- Coverage: 2023 to 2026, United States and Canada

2.2 Schema

Column	Data Type	Description
Row_ID	INT	Unique row identifier
Order_ID	VARCHAR(20)	Order identifier — multiple rows per order
Order_Date	DATE	Date the order was placed
Ship_Date	DATE	Date the order was shipped
Ship_Mode	VARCHAR(50)	Shipping class (Standard, First, Second, Same Day)
Customer_ID	VARCHAR(20)	Unique customer identifier
Customer_Name	VARCHAR(100)	Full customer name
Segment	VARCHAR(50)	Consumer, Corporate, or Home Office
Country_Region	VARCHAR(50)	Country of the order
City / State_Province	VARCHAR(50)	City and state of delivery
Region	VARCHAR(20)	Central, East, South, West
Product_ID	VARCHAR(50)	Unique product identifier
Category	VARCHAR(50)	Furniture, Office Supplies, Technology
Sub_Category	VARCHAR(50)	16 sub-categories within each category
Product_Name	TEXT	Full product name
Sales	DECIMAL(10,2)	Revenue generated from the line item
Quantity	INT	Units ordered
Discount	DECIMAL(5,2)	Discount applied (0.0 to 0.8)
Profit	DECIMAL(10,2)	Net profit on the line item

3. Data Cleaning

3.1 Steps Performed

Null Value Check

Queried all rows where Sales or Profit is NULL. Result: zero null values found. Dataset is complete on critical financial columns.

Duplicate Detection

Three levels of duplicate checking were performed:

- Full row duplicates across all 21 columns
- Duplicate Order_ID values (expected — one order has multiple products)
- Duplicate Order_ID + Product_ID combinations (true duplicates)

Finding: Two genuine full-row duplicates were identified — Order CA-2023-153623 / Product FUR-FU-10002501 (Nu-Dell Executive Frame) and Order US-2023-150119 / Product FUR-CH-10002965. Both pairs were 100% identical across all columns.

Duplicate Removal

Used a GROUP BY approach (excluding Row_ID) to preserve one row per unique business record, retaining the lower Row_ID. Final table: 10,192 rows.

Metric	Value
Original row count	10,194
Duplicates removed	2
Final row count	10,192
Null values found	0

4. SQL Analysis

All analysis was performed in MySQL 8.0 on the sales database. Queries are organized into phases, with Views created at the end for Excel export.

4.1 Analysis Phases

Phase 1 — Basic Metrics

Established headline numbers: 5,111 unique orders, \$2.33M total sales, \$292K total profit. These serve as the baseline for all subsequent analysis.

Phase 2 — Time-Series Analysis

Monthly sales and profit trends from 2023 to 2026. Key finding: strong seasonality with Sep-Dec peaks each year. Sales show consistent upward growth trend year-over-year. July 2023 was the only month with negative profit (-\$841), linked to heavy discounting.

Phase 3 — Region Analysis

Region	Total Sales	Total Profit	Margin %
West	\$739,814	\$110,799	15.0%
East	\$691,828	\$94,883	13.7%
South	\$391,722	\$46,749	11.9%
Central	\$503,171	\$39,865	7.9%

Central region has the worst margin at 7.9% despite \$503K in sales — suggesting heavy discounting or poor product mix in that region.

Phase 4 — Category & Sub-Category Analysis

Category	Sales	Profit	Margin %
Technology	\$839,893	\$146,543	17.5%
Office Supplies	\$731,893	\$126,023	17.2%
Furniture	\$754,748	\$19,730	2.6%

Furniture generates nearly as much revenue as Technology but at a fraction of the margin. Loss-making sub-categories: Tables (-\$17,753), Bookcases (-\$3,632), Supplies (-\$1,171).

Phase 5 — Discount Impact Analysis

Discount Band	Orders	Avg Profit/Order	Total Profit
No Discount (0%)	4,929	+\$66.27	+\$326,719
Low (0-20%)	3,840	+\$26.46	+\$101,599
Medium (20-40%)	464	-\$77.57	-\$35,991
High (>40%)	951	-\$105.19	-\$100,030

This is the project's most critical finding. High discounts (>40%) generated \$100,030 in total losses. The swing from no-discount (+\$66 avg profit) to high-discount (-\$105 avg profit) represents a \$171 difference per order. Discounting above 40% is destroying business value with no evidence of volume compensation.

Phase 6 — Customer RFM Analysis

Recency, Frequency, Monetary analysis performed at customer level. Each customer was scored on: how long they have been ordering (recency days), how many unique orders they placed (frequency), and their total spend (monetary). Top customers by spend exceed \$20,000 in lifetime value.

Phase 7 — Shipping Performance

Ship Mode	Orders	Sales	% of Volume
Standard Class	3,068	\$1,378,841	60.1%
Second Class	982	\$466,671	19.2%
First Class	795	\$351,751	15.6%
Same Day	266	\$129,272	5.2%

5. SQL Views Created

Eight views were created to organize analysis results and enable direct Excel import.

View Name	Contents
vw_monthly_sales_trend	Month, Total_Sales, Total_Profit, Margin_Pct — time series data
vw_region_performance	Country_Region, Region, Orders, Sales, Profit — with ROLLUP subtotals
vw_category_analysis	Category, Orders, Sales, Profit, Margin — ordered by profit
vw_subcategory_analysis	Category, Sub_Category, Sales, Profit, Margin_Pct, Avg_Discount
vw_discount_impact	Discount_Band, Orders, Avg_Profit, Total_Profit — banded analysis
vw_customer_rfm	Customer_ID, Name, Recency_Days, Total_Orders, Sales — RFM metrics
vw_shipping_analysis	Ship_Mode, Avg_Days, Orders, Sales — shipping performance
vw_loss_making_products	Product_Name, SUM(Profit) — all products with negative total profit

6. Excel Dashboard

6.1 Structure

The workbook is organized into data sheets (one per SQL view) and a final Dashboard sheet. Data sheets are the engine; the Dashboard is the only sheet visible to reviewers.

- raw_data — Full dataset from samplesuperstore.csv
- monthly_trend — From vw_monthly_sales_trend
- region — From vw_region_performance
- category — From vw_category_analysis
- discount — From vw_discount_impact
- shipping — From vw_shipping_analysis
- customers — From vw_customer_rfm
- loss_products — From vw_loss_making_products
- Pivots — Hidden sheet containing all pivot tables
- Dashboard — Final assembled view

6.2 Dashboard Components

KPI Cards (Top Row)

Four rounded-rectangle shapes with linked text boxes displaying: Total Sales (\$2.33M), Total Profit (\$292K), Profit Margin (12.6%), Average Order Value (\$464). Each card includes a subtitle line with context.

Monthly Trend Chart

Area/line chart showing Sales and Profit from Jan 2023 to Dec 2026. X-axis formatted as date axis with 3-month major intervals. Y-axis formatted as \$K currency. Reveals seasonal peaks in Sep-Dec and overall upward growth trajectory.

Region Performance Chart

Clustered bar chart comparing Sales and Profit across four regions. West leads in profit at \$110,799 (15.0% margin). Central underperforms at 7.9% margin despite \$503K in sales.

Category & Sub-Category Analysis

Horizontal bar charts showing category breakdown with margin percentage mini-cards. Highlights Furniture as the weakest category (2.6% margin) and Tables as the biggest loss-maker (-\$17,753 total profit).

Discount Impact Chart

Column chart showing average profit per order by discount band. Visual proof that discounts above 20% consistently produce negative returns. High discounts alone lost \$100,030.

Shipping Overview

Four KPI-style cards showing orders and sales by shipping mode. Standard Class dominates at 60% of all orders.

Slicers

Interactive filter buttons for Segment, Region, and Category. Connected to all pivot tables so a single slicer click filters the entire dashboard simultaneously.

7. Key Findings & Recommendations

7.1 Findings

1. **Discounts above 40% are destroying profit.** High-discount orders lost \$100,030 in total. The average profit swings from +\$66 (no discount) to -\$105 (high discount) — a \$171 difference per order with no evidence of volume compensation.
2. **Furniture has a critically low margin of 2.6%.** Despite \$754K in sales, it generates only \$19,730 in profit. Tables sub-category alone lost \$17,753.
3. **Central region underperforms significantly.** 7.9% margin vs 15.0% in the West. Likely caused by higher discount rates in that region.
4. **Strong seasonal pattern.** Sep-Dec consistently outperform Jan-Feb across all years. Q4 planning and inventory management is critical.
5. **Technology drives the most profit per dollar of sales.** 17.5% margin on \$840K in sales. Copiers (\$56K profit) and Phones (\$45K profit) are the top sub-categories.

7.2 Recommendations

- Cap discounts at 20% across all categories — especially Furniture and Tables
- Review Tables pricing strategy — \$17,753 loss is unsustainable at current discount levels
- Investigate Central region discount policies and align with West region practices
- Prioritize Technology and Office Supplies in marketing spend — highest ROI per sales dollar
- Plan Q4 inventory and staffing 60 days in advance given consistent seasonal spikes

8. Technical Stack

Tool	Version/Type	Usage
MySQL	8.0	Data storage, cleaning, all SQL analysis, views
MySQL Workbench	8.0	SQL IDE, CSV export
Microsoft Excel	365	Pivot tables, charts, interactive dashboard
CSV	UTF-8	Data transfer between MySQL and Excel

8.1 Project File Structure

```
superstore-retail-analysis/  
  data/  samplesuperstore.csv  
  sql/   superstore_analysis.sql  (all queries + views)  
  excel/ superstore_dashboard.xlsx  
  docs/  project_documentation.docx  
  README.md
```

9. About the Author

Name	Gauransh Singh (Garv)
Degree	BCA Final Year — AI/ML Specialization, Galgotias University
Skills	SQL, Python, Power BI, Tableau, Excel, Machine Learning
Certifications	IBM, AWS, Cisco, Deloitte/Forage
Achievements	Top 11 — Smart India Hackathon National Finalist — Code Astra IEEE Summit 2025
Portfolio	gauransh-singh.vercel.app
LinkedIn	linkedin.com/in/gauransh-singh-211586294
Target Roles	Data Analyst Internship — Delhi NCR (Noida/Gurgaon)

This project is part of a focused portfolio-building effort targeting data analytics internships in the Delhi NCR region, with a long-term goal of data and AI roles in Europe (Germany/Netherlands).